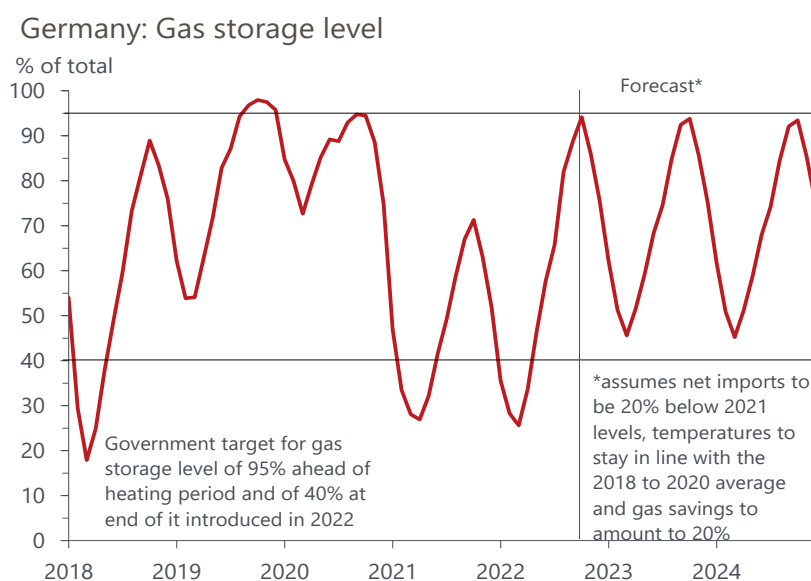


Research Briefing | Eurozone

Next winter will be tough but needn't be worse than this year's

- Replenishment of Europe's gas storages due to increased non-Russian supplies, mild weather, and gas demand reductions has eased price pressures and made gas rationing this winter less likely. While next winter could be worse, we don't think renewed recession need be the base case.
- Indeed, much of the pain has already materialised. Using Germany as an example, we show that this year's gas price spike has caused massive demand destruction, a key driver of our below-consensus GDP forecast for the eurozone. German gas use was down about 30% in October. Even adjusting for the unusually mild temperatures, consumption fell more than 20% driven by industry (-30%) and households (-21%) but partially offset by gas-based electricity production.
- We think this level of savings will be sustained through 2022 and into next year, consistent with our high gas price forecast. This would offer a positive starting point for the 2023 storage refill.
- While tight gas supply will sustain high prices, the economic costs of necessary cuts in gas use are likely to ease next year. There should be less need to burn gas to produce electricity thanks to a recovery in French nuclear power and rise in German coal-based power production. We expect more progress on fuel switching by industrial firms and think import substitution will soften the emergence of new supply bottlenecks.
- Gas supply remains the most critical variable. For our simulation, we assume German net imports will be 20% below 2021 levels, in line with recent gas flows into the EU and German net imports. This assumption accounts for Russian pipeline exports falling close to zero but not for an end of Russian LNG and not for other potential gas supply disruptions.
- A cold winter this year remains the other key risk. Temperatures 0.3° below normal would deplete storage towards critical levels requiring an additional 4% cut in gas use to refill storage to 95% ahead of next winter. But based on current trends, the weather could turn into an upside risk.

Chart 1: Under plausible assumptions, next winter will be challenging but not worse than this year



Source: Oxford Economics/Haver Analytics/Destatis/Bundesnetzagentur

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Gas supply outlook for 2023 is daunting

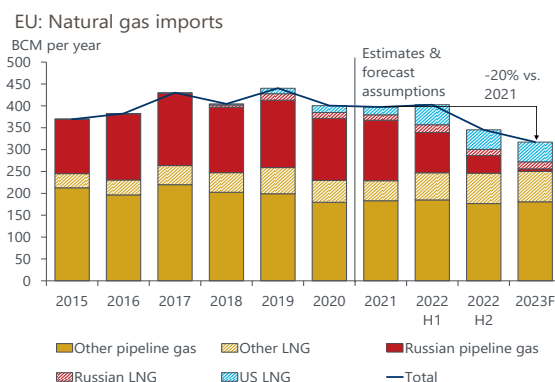
If anything, gas supply prospects for next year are even more daunting than this year. This is behind recent [warnings](#) that winter 2023 could well be worse than this year.

The reason is obvious. Russian pipeline gas exports to Europe have dropped substantially this year, and it is likely they will stop for good. And while LNG imports have partly balanced the fall in pipeline gas supply from Russia this year, global LNG export capacities are unlikely to fully offset the loss of Russian gas until 2024-2025, and LNG import demand in other regions could rebound after weakness this year. The IEA also mentioned these issues in a recent [report](#) titled "Never Too Early to Prepare for Next Winter."

So, refilling gas storage ahead of the next heating season will be challenging, and gas imports in 2023 are likely to fall short of 2022 levels, which were already down 10% to 15% compared to 2021 in recent weeks.

We will publish a more in-depth look at gas supply prospects, but we think that a fall in gas imports in 2023 of 20% versus 2021 is plausible as a baseline assumption. This is consistent with pipeline gas imports from Russia ending and imports from elsewhere remaining stable at recent levels ([Chart 2](#)). A further 5% fall is likely, should Russian LNG exports to Europe end and should the global LNG market be tighter than expected.

Chart 2: European gas imports set to fall further in 2023, with a 20% drop from 2021 plausible

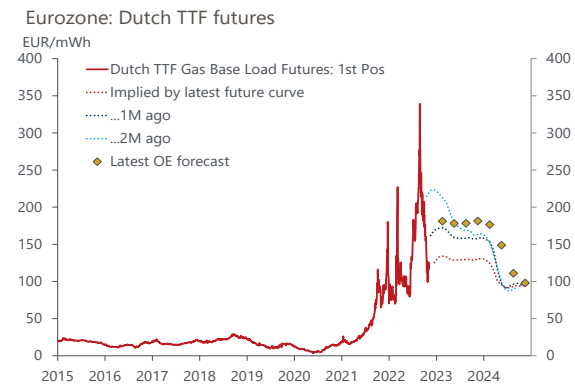


Source: Oxford Economics/Haver Analytics/Eurostat

This also means that the recent plunge in spot gas prices is unlikely to last, as Europe will need to continue to outbid other LNG importers sustaining high energy prices. This is in line with

our gas price assumptions, which are significantly above current futures prices ([Chart 3](#)).

Chart 3: Gas prices are set to stay at elevated levels throughout 2023 at least



Source: Oxford Economics/Haver Analytics

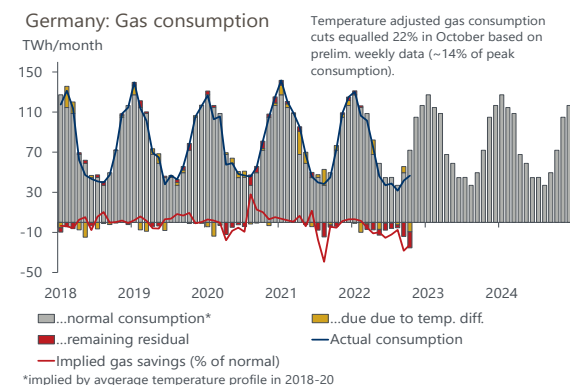
However, despite this gloomy backdrop, we think there are several factors that suggest that next year need not be worse and that a renewed recession over next winter is not inevitable.

The adjustment pain already materialised

First and foremost, this year's massive gas price spike has resulted in equally massive demand destruction. We are looking at Germany in this brief because it has timely data on gas use by sectors and because it is among the most exposed countries.

Based on preliminary data, German gas demand was down 33% y/y in October. A mild autumn drove much of that fall, but even adjusted for temperature gas use was 22% below what these temperatures would have suggested. This comes after a 28% implied saving in September ([Chart 4](#)).

Chart 4: Gas demand destruction in Germany gathered pace throughout 2022



Source: Oxford Economics/Haver Analytics

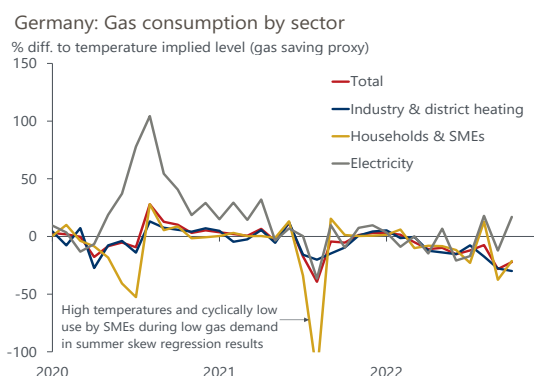
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True, it will be harder to sustain the same relative savings in peak heating season, but the demand shortfall in September and October equal about 12% and 14% of peak demand, respectively. The IEA assumed a fall in gas consumption of 11% during this peak season.

The sectoral breakdown also offers encouraging insights. Households contributed to the savings, although their prices likely respond with a larger time lag to the surge in wholesale prices that hit industry. Industrial gas consumption saw the largest fall, though.

In contrast, gas use for electricity was volatile with no clear sign of demand destruction in recent months ([Chart 5](#)). This is consistent with gas-fired power stations' function as electricity grid stabilisers, offsetting fluctuations in other power sources such as renewables. Demand for German gas-fired power production was very high this year despite sky-high gas prices as it had to partially offset the plunge in French nuclear power and the general scarcity of hydro power given drought conditions. These sectoral gas-saving numbers are also broadly consistent with a recent [working paper](#) on German data.

Chart 5: Gas savings rose throughout 2023 as prices soared



Source: Oxford Economics/Haver Analytics/NOAA/Bundesnetzagentur/Frauenhofer ISE/Destatis

There are two key implications from the fact that gas consumption has seen such a front-loaded adjustment towards levels that are about in line with the weak supply outlook for next year.

First, the sharp drop in gas consumption this year in industry means the economic pain is front-loaded. This is the main reason for our view of a sharper GDP contraction than consensus during the winter. We think this also implies less scope for next winter to be worse. After all, this data shows that the substantial amounts of Russian

gas that were still flowing in the first half of 2022 were used to refill German gas storage from a very low level, while demand was already adjusting to weaker medium-term prospects.

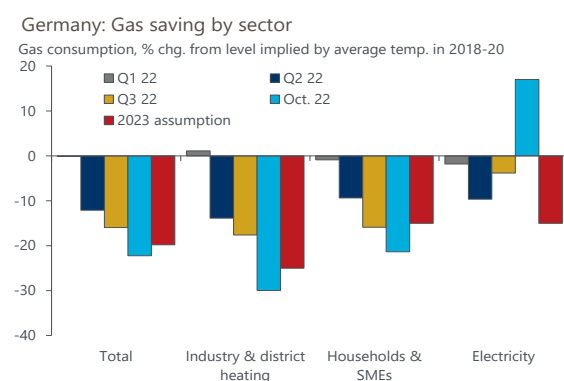
Second, the gas savings have made it more likely that Europe will get through this winter without gas rationing. And gas storage should not be too low after the winter, thus limiting the challenge to replenish stocks ahead of next winter.

Indeed, under plausible assumptions German gas storage is likely to exceed the 40% target set by the government for the end of the heating period ([Chart 1](#)).

For this, we cautiously assume that German net gas imports will be 20% below 2021 levels. We also assume temperatures will be in line with the average seen in 2018 to 2020 and that gas consumption cuts by households and the power sector reach 15% while industry needs to reach 25%. For households and industry, that is in line with recent levels.

For the power sector we think the assumed cuts are plausible, once supply issues in the European power sector ease ([Chart 6](#)). In addition, the German government set in motion plans to activate more coal-fired power stations. In any case, the power sector isn't a big swing factor for the results. But its rising contribution will lessen the pressure on industry to cut gas use.

Chart 6: Gas savings in electricity sector should rise next year



Source: Oxford Economics/Haver Analytics/NOAA/Bundesnetzagentur/Frauenhofer ISE/Destatis

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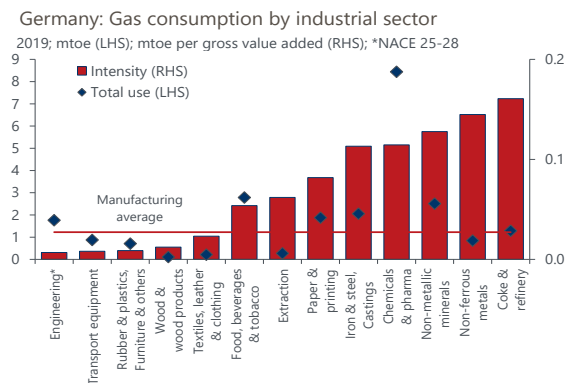
Fuel switching should ease economic drag from needed gas demand cuts

The other key factor that makes us less gloomy about next winter is that we think there is scope for fuel switching by industrial gas users to lessen the economic drag from gas demand reductions.

Admittedly, there is little in the data that would suggest a meaningful role for this yet, and we are [unlikely](#) to see much of that before sometime in 2023. Corporate decision-making takes time. There are plenty of regulatory hurdles. And supply-chain disruptions likely also delayed such initiatives.

But the article also underscores that over time gas savings and fuels switching are likely to gather pace given the huge incentive from high gas prices. There may be little scope for fuel switching in sectors that rely on gas as input material such as chemicals, but for those that use it primarily as heat source a switch to oil is plausible. These tend to be sectors with lower gas intensity and lower total use ([Chart 7](#)). But even if only those sectors were to implement any fuel switching, this could add up to gas savings of maybe 5% to 10% of industry demand.

Chart 7: Gas dependence varies a lot across industrial sectors

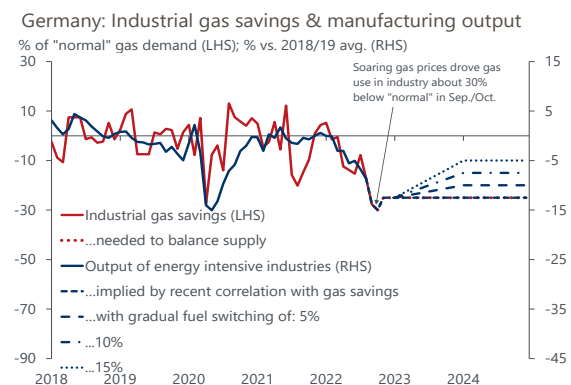


Source: Oxford Economics

This thorough [study](#), published at the onset of the war in Ukraine, highlights that the economic costs from gas consumption cuts will be materially dampened by import substitution and fuel switching even under cautious assumptions for substitution elasticities. And while the estimates may have turned out to be too optimistic for this year, we think it highly unlikely that the corporate sector will not expedite coping efforts as time goes by.

We offer a purely illustrative visualisation for this in [Chart 8](#). It shows that our gauge of realised gas demand destruction in industry closely correlated with falls in industrial activity of energy-intensive manufacturing sectors this year. The recent 30% fall in gas use corresponds to a production cut of about 15%, which also tallies with the production expectations in those sectors. If we are right on fuel switching, this correlation may weaken in future.

Chart 8: Downward pressure could ease in 2023, depending on scale of fuel switching

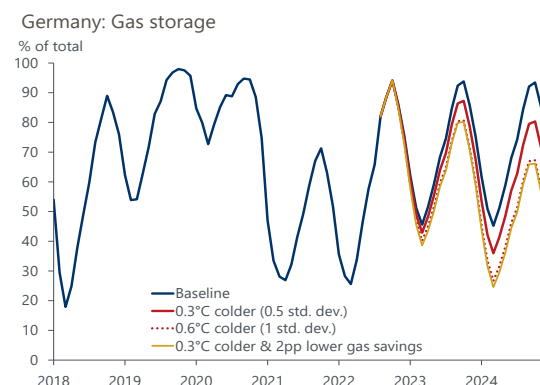


Source: Oxford Economics/Haver Analytics/Destatis/NOAA/Bundesnetzagentur

Downside risks still loom large

But there are risks to this view. For instance, while this year's autumn has been mild, winter could yet turn colder than usual. Temperatures would tighten gas markets, as was the case during the heating season 2020/2021, while a winter like in 2017/2018 would be a big threat requiring about 4ppts higher gas saving overall, or 33% instead of 25% gas saving by industry ([Chart 9](#)).

Chart 9: A colder winter or insufficient falls in gas consumption would deplete gas storage



Source: Oxford Economics/Haver Analytics

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There are downside risks for gas supply, although we think our assumptions are cautious in light of the high gas price view, which should allow Europe to continue to outbid Asia on the spot LNG market as previous longer-term supply contracts run out.

Overall, we are not trying to make a bullish case for Europe and its industry. If anything, our growth outlook for Europe and the wider world is below consensus. But we think that there is a risk of excessive pessimism on Europe's outlook for 2023/2024.

Gas demand is likely to slowly adjust to the new gas supply environment. Industry should make progress on fuel switching and energy efficiency, while likely adjusting the supply chain for the most energy-intensive products. The power sector should see better hydro energy supply, and a recovery in French nuclear power supply and more coal use should also help fill the gap until the ramp up in renewables gathers pace.

Many of our assumptions and arguments will be tested over the coming winter. But for now, we feel that some recovery in GDP growth and industrial activity in H2 next year is a sensible baseline as the economy adjusts to a new normal of structurally higher gas prices and as fiscal support measures gain traction.